



The Brain's Playback: Reconstructing Music Using Neural Decoding

By: Sahana Gopalan, Adan Sanchez, Anoushka Shah, Kishant Mohan



Introduction

Music is a core part of culture and human experience. This project replicates a groundbreaking study from the Knight Lab (UC Berkeley), where intracranial EEG (iEEG) signals were used to reconstruct the song Another Brick in the Wall, Part 1 from brain activity. Our replication provides insight into auditory processing and informs future directions for music-based BCIs.

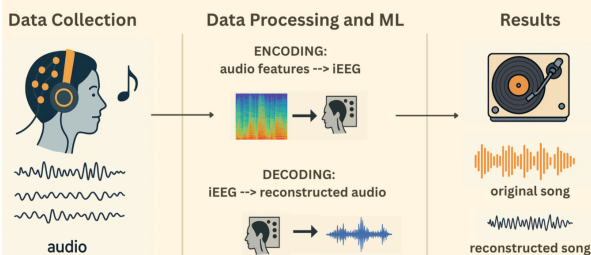
Abstract & Objectives

We replicated the Knight Lab approach for reconstructing music using iEEG data from 29 participants.

Our goals were to:

- Implement and evaluate a linear encoding/decoding model linking music features with high-frequency brain activity (HFA).
- Assess how well the model captures spectrotemporal features of the song.
- Identify methodological improvements for more accurate reconstruction (e.g., electrode selection, model architecture, and spectral analysis)

Reconstructing Music with iEEG



Materials and Methodology

We used the open-access Bellier et al. dataset, which includes intracranial EEG recordings from 29 epilepsy patients listening to Another Brick in the Wall.

1. Convert audio $\rightarrow$ auditory spectrogram.
2. Extract high-frequency activity (70–150 Hz) from ECoG/iEEG.
3. Use these features to train linear and multilayer perceptron models for:
 - Encoding: music features $\rightarrow$ brain activity
 - Decoding: brain activity (HFA) $\rightarrow$ music features
4. Evaluate reconstruction with effective R^2 .

Encoding Model

We wrote Python code to train a linear regression encoding model predicting HFA activity at each electrode from the song's auditory spectrogram features.

- Data split with GroupShuffleSplit (2-second blocks kept intact).
- Features standardized using RobustScaler.
- Trained using a single-layer Keras linear model with Huber loss and RMSprop optimizer.
- EarlyStopping used to prevent overfitting.
- Final STRF weights averaged across 250 bootstrap iterations.

This model evaluated how well each electrode's activity could be predicted directly from the song.

[github link](#)

Acknowledgements

This project was conducted through the club, Neurotech@Berkeley. We would like to sincerely thank the Knight Lab's Ludovic Bellier for lending his advice and expertise on exploring different methodologies, as well as our lead Beatrice Lowman, for her diligent support and guidance throughout this entire process.

Future Research Areas

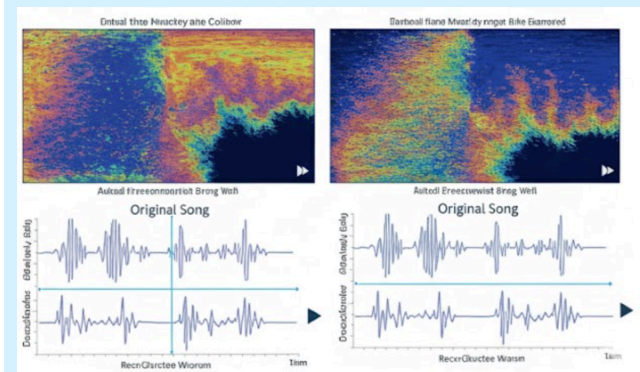
To build upon the current neural music reconstruction framework, two methodological frameworks were identified in order to examine cognitive underpinnings of music perception through novel electrode selection.

1. Vocal vs. Instrumental Electrode Selection

Comparing electrodes that respond selectively to vocals vs. instruments could reveal how the brain differentiates linguistic vs. musical content. This would help improve decoding by focusing on functionally specialized electrodes.

2. Frequency-Band Specific Encoding

Analyzing encoding performance across different EEG frequency bands (e.g., theta, alpha, gamma) may identify which neural rhythms best represent pitch, timbre, and rhythm—informing improved decoding strategies



Outreach

We met with the paper's first author, Ludovic Bellier, who advised using an 80/20 split stratified by vocals vs instruments to avoid temporal leakage and overfitting. He emphasized linear encoding for interpretability and well-tracked preprocessing as we replicate and extend the study.